Using Artificial Intelligence for Next-Generation Intelligent Transportation

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Abstract

Computer vision - emergent branch of **artificial intelligence**
- → developing algorithms
- → pass large amounts of image and video data to computers to help them make meaningful conclusions

Applying computer vision to
- → detect **pedestrian trespassing**
- → perform **object detection** of camera footage from NJ Transit railroad crossings

Current models: detect trespassing with **over 90% accuracy**,
- → near perfect accuracy is crucial to make decisions
- → alerting trains to halt because of an obstacle in the tracks
- → not enough training data for trespassing

My focus:
- → annotating images with labels so that the computer vision model can take those images as input and learn how to better detect pedestrians
- → developing a program:
  - **input**: data set of images of people in various scenarios
  - **Microsoft Common Objects in Context (COCO) data set**
  - extract the person
  - **output**: using a customized pipeline of generative models, overlays them on an image of railway tracks to simulate trespassers

Computer vison model learns **new features** and **patterns** that will help it make more **accurate** and **confident decisions**.

Background

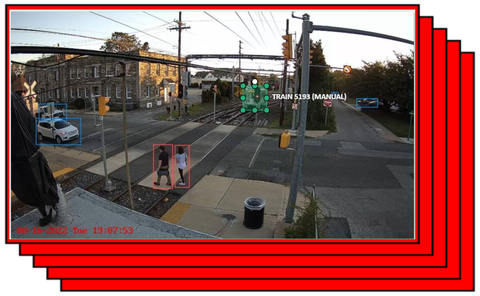
Grade crossings: intersection of railroads and roads
- → 34% of railroad incidents over the past ten years have occurred at grade crossings

Current method of a barrier + red light is not entirely effective in preventing accidents/trespassing
- → Sometimes, people will try to go around the barrier

Object detection has been leveraged for advances in autonomous driving

Materials and Methods

1. Gather + Label Data

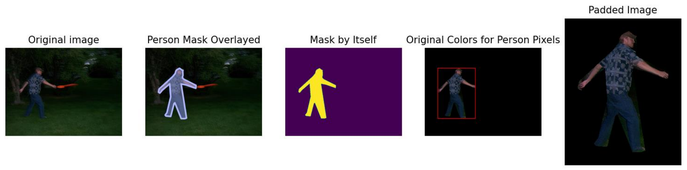


Annotated images for object detection

Labels: pedestrian, train, family vehicle

1000's of images → training data, test data

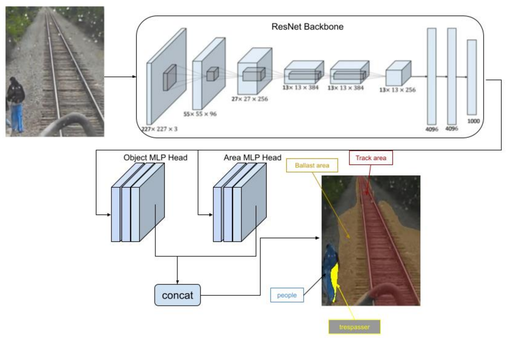
2. Generate Augmented Data Set



Parse through Microsoft COCO dataset - images of people
- → extract a mask of the person
- → overlay onto image of railway tracks

Motivation: in our data set, we do not have a lot of images of pedestrians while they are trespassing

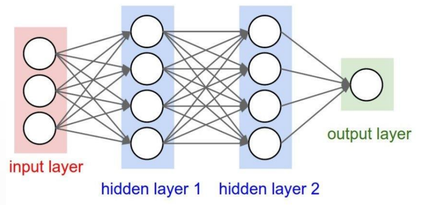
3. Build and Run the Machine Learning Model



- → Iterate through the training data
- → Identify patterns and features
- → Make predictions for test data

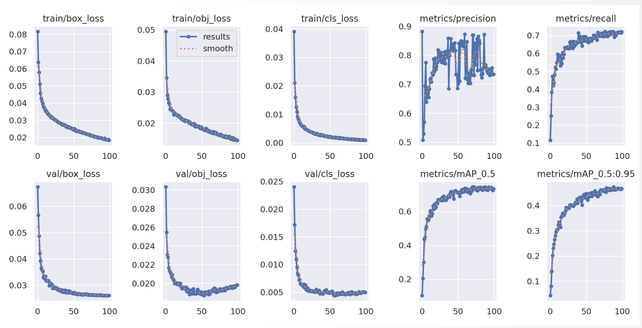
4. Improve the Model

Fine-tune **hyperparameters**:
- → Learning rate
- → Number of layers
- → Number of epochs (passes through the entire dataset)



Results

Artificially generated trespassing incident

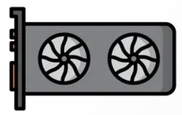


Sample Training and Test Loss Data
- → Is the model **overfitting**? It can do well on the training data set, but cannot extrapolate to the test data set

Conclusion

Computer vision can be applied to solve real world problems by leveraging a computer's resources to draw insights from image data. As developers working to improve a computer vision model, robust data that captures a variety of test cases is crucial for the model to make accurate predictions in a confident manner. When training a computer vision model, it is also important to consider the trade-offs between time, computing resources, and accuracy.

Future Direction



- ➢ Leverage newer, more advanced hardware to train models more efficiently

- ➢ Dashboard that monitors real-time trespassing events
- ➢ Gather data to identify trespassing patterns and trends – days of week, times of day

Acknowledgments

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